

PERT

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Practice quiz

A troop of young comedians is confronted with the problem of planning their next show according to various tasks listed in the table below. The troop wishes to organize the tasks as the preparation of the show lasts as short as possible. Tasks are linked together by precedence constraints, which means that some tasks cannot begin until other tasks are completed. For example, in the table below, we see that B can only begin if A is complete (A precedes B). In the same way, we see that C can only begin if A and B are terminated (A and B precede C), and so on.

Task	Description	Duration [days]	Precedence
A	Writing sketches	30	-
B	Sketches distribution	12	A
C	Choice of the place of the show	8	A and B
D	Technical cuts	4	A and C
E	Preparation of the stage set	7	C and D
F	Rehearsal of the outer sketches	10	A, B, C and D
G	Rehearsal of the inner sketches	12	D, E and F
H	Synchronization	3	F and G
I	Visual accompaniment	14	H
J	Sound accompaniment	7	I and H
K	Mixing	6	I and J
L	General rehearsal	1	K

1. Model the problem using a PERT network.
2. Identify the critical tasks that can not suffer from any delay without delaying the show, and give the minimal duration of the show preparation.

3. Calculate when each task should start at the earliest and the latest to achieve the minimal duration.